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AcademicRAG: A Knowledge Graph-Enhanced Framework for Intelligent Academic Resource Discovery

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Summary: Traditional academic retrieval systems struggle to capture complex semantic relationships; this often leads to incomplete results and missed interdisciplinary connections. We present AcademicRAG [1], a novel knowledge graph-enhanced framework that addresses these limitations through two key innovations: clue-guided keyword generation and subgraph-based retrieval. Unlike GraphRAG's expensive community structures and LightRAG's limited one-hop neighbor retrieval, AcademicRAG builds complete subgraphs while anchoring keywords in actual graph content to prevent semantic drift. Additionally, it eliminates community regeneration overhead through clue-guided keyword indexing, enabling efficient incremental updates. Evaluations across agriculture and computer science domains demonstrate that our approach consistently outperforms three state-of-the-art baselines. We demonstrate AcademicRAG's practical versatility by powering a research literature assistant that leverages clue-guided querying and subgraph navigation for precise, context-aware paper discovery, and a course discovery system that uses dynamic subgraph analysis to generate personalized learning pathways and optimize course curricula in the acoustics domain.

Keywords: Knowledge Graph, Retrieval-Augmented Generation, Graph-based RAG, Academic Resource Discovery, Subgraph Retrieval, Literature Review, Course Discovery

1. Introduction

The exponential growth of digital academic content has transformed scholarly communication, with academic databases now containing millions of papers across diverse disciplines [1]. However, this unprecedented scale creates significant challenges in knowledge discovery and synthesis, particularly in academic contexts where understanding depends on grasping complex relationships between entities, methodologies, and concepts [2].

Traditional retrieval systems demonstrate fundamental limitations in academic knowledge discovery. Sparse retrieval methods like BM25, while computationally efficient, struggle with terminology variations and fail to capture semantic relationships inherent in academic knowledge [3]. Dense retrieval methods using Transformer-based encoders improve semantic understanding but cannot preserve the intricate relational structures that define academic knowledge hierarchies, prerequisite chains, and interdisciplinary connections [4].

Recent Retrieval-Augmented Generation (RAG) approaches, though promising, inherit these structural limitations. Conventional RAG treats academic documents as isolated entities with flat vector representations, failing to model the complex relationship networks essential for comprehensive academic understanding [5]. Four main RAG paradigms have emerged—Sequential, Branching, Conditional, and Loop RAG [6]—but none adequately addresses the spectrum of academic information needs.

Graph-based RAG architectures attempt to address these limitations but face critical trade-offs [7]. GraphRAG [8] employs hierarchical community structures, achieving improvement in critical dimensions over traditional RAG, but requires high computational complexity for community generation, making incremental updates prohibitively expensive. LightRAG [9] reduces computational overhead through dual-level retrieval but limits relationship discovery to one-hop neighbors, fundamentally constraining multi-hop dependency capture crucial for academic synthesis. Other frameworks (GRAG [10], HybridRAG [11], CG-RAG [12]) face similar challenges: high computational requirements, limited multi-hop inference capabilities, and manual knowledge graph curation needs.

[1] Code and additional implementation details are available at: https://github.com/shua-chen/academicRAG

We present AcademicRAG, a knowledge graph-enhanced RAG framework specifically designed for intelligent academic resource discovery. Our approach introduces two key technical innovations that address the fundamental limitations of existing graph-based RAG frameworks. Firstly, we develop clue-guided keyword generation that anchors keyword extraction processes in actual graph content, thereby preventing semantic drift and hallucination issues that commonly plague generative retrieval systems when operating without grounded constraints. Secondly, we implement subgraph-based retrieval mechanisms that construct complete relationship networks efficiently, eliminating the need for computationally expensive community structures while avoiding the restrictive single-hop neighbor limitations that constrain existing approaches.

Our contributions are threefold. First, we develop a dual-flow architecture transforming unstructured academic content into queryable knowledge graphs through iterative entity-relationship extraction. Second, we introduce efficient algorithms for local subgraph construction and global relationship discovery that capture multi-hop dependencies while enabling incremental updates, validated through substantive evaluation. Third, we demonstrate practical versatility through two real-world applications addressing distinct academic information needs across different user groups.

Comprehensive technical details, implementation specifics, and extended case studies are available in the complete documentation [13].

2. AcademicRAG Framework

AcademicRAG operates through two primary flows: Index Flow for knowledge graph construction and Query Flow for information retrieval, as illustrated in Figure 1. This dual-flow architecture transforms unstructured academic content into structured knowledge representations while enabling sophisticated query processing that captures both local detail and global context.

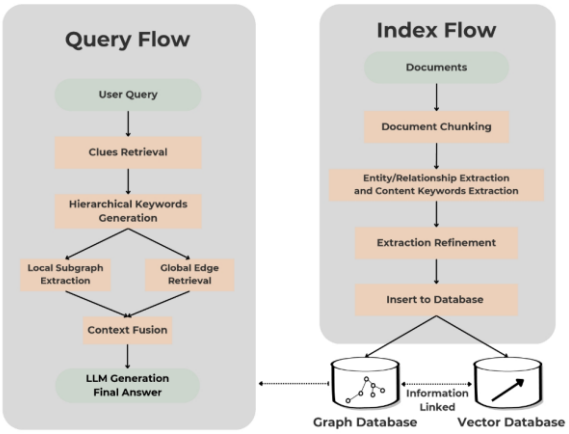

Fig. 1. AcademicRAG Framework Flowchart.

2.1. Index Flow

The Index Flow processes academic documents through a comprehensive multi-stage pipeline designed to preserve semantic relationships while enabling efficient retrieval. The process begins with document ingestion and structure-aware chunking that respects document organization, particularly important for academic content with inherent hierarchical structure such as research papers and course syllabi.

Each document chunk undergoes entity and relationship extraction using domain-adapted LLMs for academic content. The extraction process simultaneously identifies three critical components:

$$(E_i, R_i, K_i) = \mathrm{LLM}(c_i) \tag{1}$$

where c_i represents document chunks, E_i and R_i are extracted entities and relationships, and K_i are content keywords capturing global themes.

Next, the framework employs a multi-pass iterative refinement mechanism that maximizes entity and relationship extraction accuracy. Unlike single-pass extraction methods, this approach performs systematic re-extraction to capture complex academic relationships that may be missed in initial processing.

The extracted information is stored in a dual-database architecture: the Graph Database preserving structural relationships and the Vector Database storing semantic embeddings. This hybrid approach enables both structured traversal and similarity-based retrieval, providing comprehensive coverage for complex academic queries.

2.2. Query Flow

Upon receiving a user query Q, the system implements a sophisticated multi-stage retrieval strategy that addresses semantic drift issues common in knowledge graph querying. The process begins with clue-guided keyword generation, which anchors keyword extraction in actual database content to prevent hallucination:

$$K^{\mathrm{content}} = \{\, k \in K \mid \mathrm{sim}(Q, k) \geq \tau \,\}, \tag{2}$$

where K is the set of indexed content keywords, Q is the user query, and τ is a similarity threshold. The function $\mathrm{sim}(Q, k)$ computes the embedding cosine similarity between query and keyword. These clues K^{content}, anchored in actual database content, guide the LLM to generate hierarchical keywords that avoid semantic drift by combining semantic information from the user query with natural language instructions:

$$(K_h, K_l) = \mathrm{LLM}(Q, K^{\mathrm{content}}), \tag{3}$$

where K_h represents high-level conceptual keywords and K_l represents low-level contextual keywords. Low-level keywords drive local subgraph

extraction through the structured process detailed in Figure 2. Unlike LightRAG's [9] one-hop limitation, our approach constructs complete subgraphs that capture multi-hop relationships between specific entities, enhancing retrieval depth and knowledge integration.

Algorithm 1 Local Information Extraction based on Subgraph

Require: Low-level keywords $\mathcal{K}_l$, Knowledge graph $\mathcal{G} = (V, E)$
Ensure: Local entities E_{local}, relationships R_{local}, text units T_{local}
Node Matching: Match graph nodes against $\mathcal{K}_l$ using vector similarity metrics
$V_{\text{matched}} \leftarrow \{v \in V \mid \max_{k \in \mathcal{K}_l} \text{sim}(v, k) \geq \theta_{\text{node}}\}$
Subgraph Construction: Generate subgraph by finding shortest paths between matched nodes
$\mathcal{G}_{\text{sub}} \leftarrow \text{ShortestPaths}(V_{\text{matched}}, \mathcal{G})$
Pruning: Filter edges based on semantic relevance thresholds
$E_{\text{filtered}} \leftarrow \{e \in E_{\text{sub}} \mid \max_{k \in \mathcal{K}_l} \text{sim}(e.\text{description}, k) \geq \theta_{\text{edge}}\}$
Remove isolated nodes after edge filtering
Text Unit Extraction: Extract and rank textual chunks from retained edges
$T_{\text{chunks}} \leftarrow \text{ExtractTextUnits}(E_{\text{filtered}})$
Rank T_{chunks} by frequency within subgraph
Edge Expansion: Iteratively expand subgraph with one-hop neighbors
$E_{\text{expanded}} \leftarrow E_{\text{filtered}} \cup \text{OneHopNeighbors}(V_{\text{matched}}, \theta_{\text{relevance}})$
Data Truncation: Truncate all data to meet computational constraints
while $\text{TokenCount}(E_{\text{local}}, R_{\text{local}}, T_{\text{local}}) > \tau_{\max}$ **do**
  Remove lowest-ranked elements
end while
return $(E_{\text{local}}, R_{\text{local}}, T_{\text{local}})$

Fig. 2. Local Information Extraction based on Subgraph

High-level keywords enable global relationship discovery through the systematic process illustrated in Figure 3. The framework queries the vector database to identify edges correlating with broader conceptual frameworks, eliminating GraphRAG's [8] community structure overhead while maintaining comprehensive global knowledge discovery.

Algorithm 2 Global Information Extraction

Require: High-level keywords $\mathcal{K}_h$, Knowledge graph $\mathcal{G} = (V, E)$, Vector database $\mathcal{D}_{\text{vector}}$
Ensure: Global entities E_{global}, relationships R_{global}, text units T_{global}
Edge Retrieval: Query vector database for edges correlating with $\mathcal{K}_h$
$E_{\text{candidate}} \leftarrow \{e \in E \mid \max_{k \in \mathcal{K}_h} \text{sim}(\text{embed}(e), \text{embed}(k)) \geq \theta_{\text{global}}\}$
Prioritize edges encapsulating conceptual relationships
Node Retrieval: Extract nodes related to retrieved edges
$V_{\text{related}} \leftarrow \{v \in V \mid \exists e \in E_{\text{candidate}}, v \in \{e.\text{source}, e.\text{target}\}\}$
Sort V_{related} by node degree
Edge Data Processing: Analyze edge metadata
for $e \in E_{\text{candidate}}$ **do**
  $e.\text{degree} \leftarrow \text{degree}(e.\text{source}) + \text{degree}(e.\text{target})$
  $e.\text{weight} \leftarrow \text{SemanticConnectionStrength}(e)$
end for
Integration and Ranking: Rank edges using composite metric
$\text{score}(e) \leftarrow \alpha \cdot \text{RelevanceScore}(e, \mathcal{K}_h) + \beta \cdot e.\text{weight}$
$E_{\text{ranked}} \leftarrow \text{Sort}(E_{\text{candidate}}, \text{score})$
Text Unit Extraction: Extract text chunks associated with edges
$T_{\text{chunks}} \leftarrow \text{ExtractTextUnits}(E_{\text{ranked}})$
Rank by similarity between edges and $\mathcal{K}_h$
Data Truncation: Truncate to meet computational constraints
while $\text{TokenCount}(E_{\text{global}}, R_{\text{global}}, T_{\text{global}}) > \tau_{\max}$ **do**
  Remove lowest-ranked elements
end while
return $(E_{\text{global}}, R_{\text{global}}, T_{\text{global}})$

Fig. 3. Global Information Extraction

The final synthesis stage combines retrieved information through sophisticated contextual fusion to generate comprehensive responses. The system integrates locally extracted subgraph information with globally retrieved relationship networks, merging entities, relationships, and supporting textual evidence into a unified contextual framework. This integrated information is then embedded within a domain-specific prompt template designed for academic content processing. The large language model processes this enriched context to generate the final answer, leveraging both the granular detail from local subgraphs and the broader conceptual understanding from global edge networks. This fusion approach ensures that responses maintain semantic coherence while incorporating multiple levels of contextual depth, from specific entity relationships to overarching domain knowledge.

3. Evaluation and Results

3.1. Experiment Settings

We evaluated AcademicRAG using a subset of the UltraDomain benchmark [14], selecting agriculture and computer science domains for their distinct knowledge structures and citation patterns. Following Microsoft's GraphRAG methodology [8], we generated five synthetic academic personas per domain (e.g., "PhD Student in NLP," "Organic Farming Consultant"), each assigned 25 diverse queries reflecting common academic information needs, resulting in 125 evaluation queries per domain.

We compared AcademicRAG against three representative baselines: NaiveRAG [5], GraphRAG [8] and LightRAG [9]. These baselines were selected to provide comprehensive coverage of current RAG paradigms, from flat document processing to sophisticated graph-based approaches.

All frameworks used identical configurations with Qwen2.5-72B [15] for both knowledge graph construction and response generation, ensuring fair comparison. Evaluation employed DeepSeek-R1 [16] as an LLM judge conducting pairwise comparisons across four dimensions:

-**Comprehensiveness**: Coverage of relevant information and depth of analysis

-**Diversity**: Variety of perspectives and information sources

-**Empowerment**: Usefulness for decision-making and further research

-**Overall**: Holistic assessment integrating all dimensions for comprehensive performance.

To mitigate positional bias, we alternated answer placement across three trials and computed averaged win rates. While LLM-as-Judge represents current best practice for high-level semantic evaluation, we acknowledge the need for human expert validation in future work.

3.2. Evaluation Results

As shown in Table 1, AcademicRAG demonstrated consistent superiority across both domains and all evaluation dimensions. Our framework achieved overall win rates of 57.2%, 54.8%, and 52.4% against NaiveRAG, LightRAG, and GraphRAG respectively in agriculture, with even more pronounced advantages in computer science (77.5%, 56.6%, and 53.6%). The significant performance disparity between domains

suggests that field-specific characteristics—including terminology standardization and conceptual hierarchy—influence framework effectiveness, with computer science's more structured knowledge representation facilitating superior retrieval quality.

The results validate the core technical innovations of our dual-level retrieval approach. AcademicRAG's subgraph-based retrieval enables extraction of richer, more structured information compared to flat document approaches, while clue-guided keyword generation ensures semantic alignment with indexed content, preventing hallucination issues in keyword generation. Notably, while consistently outperforming LightRAG across all dimensions, the comparison with GraphRAG reveals that our framework excels in comprehensiveness and empowerment—the most critical metrics for academic information discovery—demonstrating that complete subgraph construction captures essential multi-hop relationships that community-based approaches may overlook.

Table 1. Comparative Win Rates (%) of AcademicRAG Against Baseline Models Across Two Domains

Framework	Agriculture				Computer Science			
	Comp	**Div**	**Emp**	**Overall**	**Comp**	**Div**	**Emp**	**Overall**
NaiveRAG	62.8%	58.0%	51.6%	**57.2%**	67.1%	83.2%	75.7%	**77.5%**
LightRAG	50.0%	55.2%	58.4%	**54.8%**	56.2%	57.8%	59.8%	**56.6%**
GraphRAG	52.4%	50.8%	54.0%	**52.4%**	51.2%	46.8%	55.2%	**53.6%**

Note: Comp = Comprehensiveness, Div = Diversity, Emp = Empowerment. Win rates were computed based on pairwise comparisons conducted by a Large Language Model (LLM-as-Judge) over 125 domain-specific queries, following common practices in graph-based RAG framework evaluation [8,9].

Table 2. Ablation Studies on Win Rates (%) Compared with GraphRAG

Settings	Agriculture				Computer Science			
	Comp	**Div**	**Emp**	**Overall**	**Comp**	**Div**	**Emp**	**Overall**
AcademicRAG	52.4%	50.8%	54.0%	52.4%	51.2%	46.8%	55.2%	53.6%
-clues	53.6%	48.4%	60.8%	57.2%	51.0%	43.0%	54.6%	51.8%
-subgraph	50.8%	51.6%	52.8%	53.6%	49.2%	48.0%	59.3%	55.2%

Note: Comp = Comprehensiveness, Div = Diversity, Emp = Empowerment. "-clues" and "-subgraph" represent ablated versions removing the respective retrieval mechanisms, with "-subgraph" using one-hop retrieval only.

3.3. Ablation Study

To validate our technical innovations, we conducted ablation studies examining the individual contributions of clue-guided keyword generation and subgraph-based retrieval. Table 2 presents the results of removing each component against GraphRAG as baseline. Removing clues resulted in mixed performance changes, with diversity scores declining from 50.8% to 48.4% in agriculture and from 46.8% to 43.0% in computer science, while surprisingly showing improvements in empowerment (54.0% to 60.8% in agriculture). The subgraph removal variant demonstrated similar complexity, with reduced comprehensiveness in computer science (51.2% to 49.2%) but improved overall performance in the same domain (53.6% to 55.2%).

These mixed ablation results reflect the inherent complexity of graph-based RAG evaluation and the nuanced trade-offs between our technical components. The unexpected improvements in certain configurations can be attributed to several factors. Firstly, the subjective nature of LLM-as-Judge evaluation may lead to scenarios where simpler responses are occasionally preferred for specific query types, potentially masking the true effectiveness of more sophisticated retrieval mechanisms. Secondly, the interaction effects between components create non-linear performance relationships, wherein the removal of one component may inadvertently optimize the remaining system configuration for particular evaluation criteria. Thirdly, GraphRAG's community-based approach introduces variable noise levels that differentially affect comparisons, as the quality and coherence of generated communities can vary significantly across different knowledge domains and corpus structures.

4. Applications

To demonstrate AcademicRAG's practical versatility beyond the evaluated domains, we developed two real-world applications that address distinct academic information needs across different user groups while validating the framework's adaptability to diverse academic contexts.

4.1. Course Discovery System

We adapted AcademicRAG for educational content navigation using 29 acoustics-related course syllabi from KTH Royal Institute of Technology. The system required minimal framework modifications, including education-specific entity schemas for courses and prerequisites, course-structure-preserving chunking strategies, and domain-tailored prompts optimized for educational content processing.

The resulting knowledge graph comprised 1,211 nodes and 2,252 edges with strong local clustering, demonstrating effective capture of course relationships and prerequisite structures. The largest connected component encompassed 94.63% of all nodes, indicating well-integrated knowledge representation across the curriculum.

Case study evaluation focused on two distinct user groups with different information needs. For students, the system demonstrated sophisticated curriculum mapping capabilities by generating logical learning progressions that respect educational hierarchies, providing precise prerequisite identification for advanced courses, and delivering multi-dimensional course recommendations balancing theoretical foundations with practical laboratory components. For faculty and administrators, the system successfully classified all courses by academic level and offering term, identified content overlaps through learning outcome analysis, and generated specific integration proposals with clear implementation methodologies. These capabilities enable evidence-based curriculum optimization and resource allocation decisions that transform abstract analyses into executable administrative workflows.

4.2. Research Literature Assistant

The Research Literature Assistant was developed using over 80 research papers across multiple academic domains, with framework adaptations including specialized prompts for academic content, automated title and introduction extraction methods, and structure-aware chunking strategies that preserve paper organization.

We conducted comprehensive evaluation through literature review scenario testing, focusing on blind face restoration within computer vision as a representative domain with methodological diversity. The assistant demonstrated exceptional performance across multiple dimensions: identifying major research trends and specific gaps in the field, providing contextual paper recommendations with multi-faceted contribution analysis, extracting detailed technical information typically requiring manual paper review, and organizing literature into coherent methodological categories spanning fundamental works, domain-specific applications, and hybrid approaches.

User testing with graduate students revealed significant advantages over online LLM-based systems, particularly in accommodating larger paper corpora while maintaining higher factual accuracy. The system's grounding in indexed content minimized probability of hallucination issues common in general-purpose LLMs, ensuring responses remained anchored to actual paper content. Key strengths included effective research gap identification, precise semantic search capabilities across multiple papers, and comprehensive paper relationship mapping that revealed hidden connections between publications.

4.3. Practical Insights and Limitations

Both applications demonstrated the framework's cross-domain adaptability through its modular design and LLM-driven entity-relationship extraction with domain-specific prompt engineering. However, several practical limitations emerged that warrant acknowledgment. Firstly, the framework exhibits weaker comprehension of non-textual elements such as figures, mathematical formulas, and complex diagrams, which constitute critical components of academic content across many disciplines. Secondly, the system demonstrates significant dependency on underlying LLM capabilities, with quality variations in the base model directly impacting system reliability and output consistency. Thirdly, our experimental validation was constrained by computational resource limitations, restricting the scale of evaluation to relatively small-scale deployments, which may limit the generalizability of findings to large-scale institutional applications. Despite these constraints, both applications provided substantial value for academic workflows, particularly in educational pathway planning and literature synthesis tasks, validating AcademicRAG's potential for transforming academic resource discovery across diverse institutional contexts while highlighting areas requiring future research attention.

5. Conclusions

This work presents AcademicRAG, a novel knowledge graph-enhanced RAG framework that significantly advances academic resource discovery through two key technical innovations: clue-guided keyword generation and subgraph-based retrieval. By anchoring keyword extraction in actual graph content and constructing complete subgraphs, our framework addresses fundamental limitations in existing approaches while achieving superior computational efficiency compared to GraphRAG's community-

based architecture. Comprehensive evaluation across agriculture and computer science domains demonstrates consistent performance advantages over state-of-the-art baselines, with win rates exceeding 50% across all evaluation dimensions. The framework's dual-level retrieval approach effectively balances local detail extraction with global context discovery, particularly excelling in comprehensiveness and user empowerment—the most critical metrics for academic information discovery.

The practical utility of AcademicRAG is validated through successful deployment of two real-world applications: a Course Discovery System enabling sophisticated curriculum analysis, and a Research Literature Assistant providing factually grounded literature synthesis capabilities. Both applications demonstrate the framework's versatility and cross-domain adaptability through minimal modifications, successfully addressing distinct information needs for students, faculty, and researchers. The applications revealed important insights about academic workflows while highlighting the framework's ability to transform unstructured academic content into semantically rich, queryable knowledge representations that support evidence-based decision-making.

Despite these advances, several limitations warrant acknowledgment, including significant dependency on underlying LLM capabilities, weaker comprehension of non-textual elements, and restricted experimental validation scale. Future work should prioritize multi-modal content processing, scalability optimization, and comprehensive human expert evaluation to strengthen the foundation for next-generation academic resource discovery systems.

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References

[1]. W. Ammar et al., Construction of the Literature Graph in Semantic Scholar, *in Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 3 (Industry Papers), 2018, pp. 84–91.

[2]. S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu, A survey on knowledge graphs: Representation, acquisition, and applications, *IEEE transactions on neural networks and learning systems*, vol. 33, no. 2, Feb. 2022, pp. 494–514.

[3]. S. Robertson and H. Zaragoza, The probabilistic relevance framework: BM25 and beyond, *Foundations and Trends® in Information Retrieval*, vol. 3, no. 4, 2009, pp. 333–389.

[4]. J. Wang et al., Utilizing bert for information retrieval: Survey, applications, resources, and challenges. *ACM Computing Surveys*, vol. 56, no. 7, 2024, pp. 1-33.

[5]. P. Lewis et al., Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, *33*, 2020, pp.9459-9474.

[6]. X. Li et al., From matching to generation: A survey on generative information retrieval, *ACM Transactions on Information Systems*, vol. 43, no. 3, 2025, article 83.

[7]. B. Peng et al., Graph retrieval-augmented generation: A survey, *arXiv preprint arXiv:2408.08921*, 2024.

[8]. D. Edge et al., From local to global: A graph RAG approach to query-focused summarization, *arXiv preprint arXiv:2404.16130*, 2024.

[9]. Z. Guo, L. Xia, Y. Yu, T. Ao, and C. Huang, LightRAG: Simple and fast retrieval-augmented generation, *arXiv preprint arXiv:2410.05779*, 2024.

[10]. Y. Hu, Z. Lei, Z. Zhang, B. Pan, C. Ling, and L. Zhao, GRAG: Graph retrieval-augmented generation, *arXiv preprint arXiv:2405.16506*, 2025.

[11]. B. Sarmah et al., HybridRAG: Integrating knowledge graphs and vector retrieval augmented generation for efficient information extraction, *In Proceedings of the 5th ACM International Conference on AI in Finance*, 2024, pp. 608-616.

[12]. Y. Hu, Z. Lei, Z. Dai, A. Zhang, A. Angirekula, Z. Zhang, and L. Zhao, CG-RAG: Research question answering by citation graph retrieval-augmented LLMs, *arXiv preprint arXiv:2501.15067*, 2025.

[13]. S. Chen and Z. Liu, AcademicRAG: Knowledge Graph Enhanced Retrieval-Augmented Generation for Academic Resource Discovery, KTH: urn:nbn:se:kth:diva-364916, Aalto: urn:nbn:fi:aalto-202506164710, 2025.

[14]. H. Qian, P. Zhang, Z. Liu, K. Mao, and Z. Dou, MemoRAG: Moving towards next-gen RAG via memory-inspired knowledge discovery, *arXiv preprint arXiv:2409.05591*, 2025.

[15]. A. Yang et al., Qwen2.5 technical report, *arXiv preprint arXiv:2412.15115*, 2024.

[16]. DeepSeek-AI, DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, *arXiv preprint arXiv:2501.12948*, 2025.